

CSMFAB Materials Study — Part IV

Polymers, Electronics, Power Systems, and Support Materials

Carrington Storm Motors | Safe Pod Engineering Company | June 2026

Study Scope: Engineering polymers (PEEK, PEX-a, PEN, PPSU), functional electronics (GaN RF, LoRa, LiFePO4), soapstone, UHMWPE rope hardware, magnetorheological elastomers, optical fiber systems, and construction support materials.

Material 22 — PEEK (Polyether Ether Ketone) and PEEK CF40

Formula: Repeating units of ether-ether-ketone groups | **Class:** High-performance semi-crystalline thermoplastic

22.1 Physical Properties

- Tensile strength (neat): 100 MPa; CF40 composite: 220 MPa
- Continuous service temperature: 260°C
- Chemical resistance: excellent (all solvents, acids except concentrated H2SO4)
- Electrical resistivity: $>10^{15} \Omega \cdot m$ (neat); $>10^{10} \Omega \cdot m$ (CF40 — carbon fiber just below percolation)
- Thermal conductivity: 0.25 W/m·K (low — thermal isolator)
- Biocompatibility: FDA-approved for implants

22.2 How PEEK is Made

Monomer: 4,4'-difluorobenzophenone + hydroquinone **Synthesis:** Nucleophilic aromatic substitution in diphenyl sulfone solvent □ PEEK polymer **Raw materials:** - Benzene (from petroleum refining): \$0.80/kg - HF gas: \$1.50/kg (fluorination step) - Hydroquinone: \$3.50–\$6.00/kg (phenolic chemistry) - K2CO3: \$0.80/kg (base)

Primary producers: - Victrex plc (UK) — dominant, ~70% global market share - Evonik Vestakeep (Germany) - Solvay KetaSpire (Belgium) - Jilin Zhongyan (China)

22.3 Commercial Pricing

Grade	Price/kg (2026)	Application
PEEK granules (neat)	\$90–\$140/kg	Machining, molding
PEEK rod/bar stock	\$140–\$210/kg	Machined parts
PEEK CF40 (40% CF)	\$220–\$320/kg	Structural composite
PEEK CF40 rod	\$280–\$380/kg	Machined structural
PEEK GF30 (30% GF)	\$150–\$220/kg	Less conductive CF replacement
PEEK sheet	\$180–\$260/kg	Flat structural

22.4 In-House Production: NOT Recommended

PEEK synthesis requires specialized ether coupling reactors (\$2M+) and diphenyl sulfone handling. **Strategy: Purchase commercial PEEK and machine/mold in-house.**

In-house PEEK machining: - CNC lathe/mill with carbide tooling: \$25,000–\$65,000 - PEEK machines cleanly with conventional tooling - Machined part cost: material + \$8–\$25/part machining

In-house injection molding of PEEK components: - PEEK injection molder (temperature-capable to 380°C): \$45,000–\$80,000 - Mold tooling: \$3,000–\$15,000 per design (steel mold; PEEK is abrasive) - This reduces PEEK part cost by 40-60% vs. buying machined parts from suppliers

Feasibility Rating: 3/5 for in-house manufacturing (buy material, add value via machining/molding)

Purchasing strategy: Establish volume supply agreement with Victrex (minimum 500 kg/yr commitment gets 15-25% price reduction).

Products: 006, 010, 012, 013, 016, 017, 019, 020, 024, 026, 028, 031, 033, 035, 038, CSMMECH 001–003, CSMIC 105, 106

Material 23 — PEX-a (Cross-linked Polyethylene Pipe)

Class: HDPE cross-linked via peroxide method (a-type) | **Application:** Non-conductive water distribution

23.1 Properties

- Electrical resistivity: $>10^{12} \Omega \cdot m$ (complete GIC immunity for water piping)
- Service temperature: 95°C continuous (ASTM F876)
- Working pressure: 160 psi at 23°C; 80 psi at 82°C
- Memory effect: coils/bends return to shape at elevated temp — installation advantage
- Zero corrosion, -40°C freeze survivability (flexible when frozen)

23.2 Raw Material: Polyethylene

Ethylene source: - Steam cracking of naphtha/ethane (petroleum refining by-product) - Bio-ethylene: from bioethanol dehydration (Braskem — sugarcane feedstock, Brazil/Texas) - HDPE price: \$1.20–\$1.80/kg (commodity — no in-house benefit)

Cross-linking (Engel peroxide method for PEX-a): - HDPE + organic peroxide (dicumyl peroxide 0.5%)
□ extruded □ heated chambers □ cross-linking - Degree of cross-link: >70% (ASTM F876 requirement)

23.3 Commercial PEX-a Pricing

Size	Price/ft (USD, 2026)	Price/m
1/2" PEX-a	\$0.55–\$0.75/ft	\$1.80–\$2.45/m
3/4" PEX-a	\$0.85–\$1.15/ft	\$2.80–\$3.75/m
1" PEX-a	\$1.40–\$1.90/ft	\$4.60–\$6.20/m
1.5" PEX-a	\$2.20–\$3.00/ft	\$7.20–\$9.80/m

Domestic suppliers: Uponor (Minnesota), Rehau (USA operations), Viega, Watts Radiant

In-house production: PEX-a extrusion requires \$150,000–\$400,000 extrusion line. At current low pricing, commercial purchase is superior unless volume exceeds 100 km/yr. **Feasibility Rating:** 2/5 — buy commercial.

Products: 012 (plumbing system)

Material 24 — PEN (Polyethylene Naphthalate) — Non-Conductive Spray Cans

Class: Aromatic polyester | **Key advantage over PET:** Better gas barrier, higher Tg (120°C vs. 80°C)

24.1 Properties

- Tensile strength: 60–90 MPa
- Oxygen barrier: 5× better than PET
- CO₂ retention: ideal for pressurized aerosol cans
- Completely non-conductive: $>10^{14} \Omega\cdot\text{m}$
- Recyclable thermoplastic (melt recycling)

24.2 Raw Materials

Naphthalene dicarboxylic acid (NDC): From naphthalene (coal tar processing or trimethylnaphthalene oxidation) - NDC price: \$4–\$7/kg **Ethylene glycol (EG):** byproduct of ethylene oxide hydration; \$0.60–\$1.00/kg

Polymerization: NDC + EG $\rightarrow$ PEN polyester (similar to PET production but different monomer) PEN pellets: \$3.50–\$6.50/kg (vs. PET \$1.20–\$1.80/kg — 2.5–3× premium for better performance)

24.3 Blow Molding to Bottles/Cans

- Injection stretch blow molding (ISBM) — same as PET bottle production
- Suitable equipment: standard PET ISBM machine with temperature adjustment
- Equipment cost: \$45,000–\$120,000 (1-cavity to 4-cavity)
- PEN requires ~20°C higher processing temperature vs. PET

In-house cost: PEN pellets \$4.50/kg + ISBM cycle cost \$0.12/bottle = **\$0.65–\$0.95 per 150mL can body** vs. commercial steel aerosol can: \$0.45–\$0.75 each (only for commodity volumes of millions)

Feasibility Rating: 4/5 — using existing ISBM infrastructure (or modest capital investment)

Products: 018 (non-conductive spray cans)

Material 25 — Soapstone / Steatite ($\text{Mg}_3\text{Si}_4\text{O}_{10}(\text{OH})_2$)

Mineral class: Talc/steatite hydrous magnesium silicate | **Application:** Thermal mass, cooking, heating

25.1 Physical Properties

- Thermal mass (volumetric heat capacity): 2.22 MJ/m³·K
- Thermal conductivity: 6.5 W/m·K (spreads heat evenly — no hot spots)
- Max service temperature: 1000°C (no phase change below)
- Machinability: excellent with standard tooling (soapstone is the softest stone used industrially)
- Completely non-conductive: $>10^{12} \Omega\cdot\text{m}$
- Emissivity: 0.91 (excellent FIR emitter for cooking)

25.2 Mining and Acquisition

Primary sources: - **India (#1):** Rajasthan, Uttarakhand — largest global producer - **China (#2):** Liaoning province - **Brazil:** Minas Gerais - **USA domestic:** Vermont (soapstone quarries active), Virginia, Massachusetts - **Finland:** Tulikivi brand (soapstone fireplaces/ovens — proven market)

Cost at mine/quarry: - Raw soapstone block: \$2–\$5/kg (India/China); \$8–\$15/kg (Vermont domestic) - Soapstone slab (surface-finished): \$40–\$80/linear ft (architectural market)

25.3 Machining and Shaping

Soapstone machines beautifully: - Standard woodworking tools work (so soft — Mohs 1-2) - CNC waterjet cutting for precise profiles: \$1–\$4/linear inch - Hand-shaping: \$15–\$35/hour labor

In-house soapstone processing: - Angle grinders, diamond saw blades, belt sanders: \$2,000–\$8,000 total equipment - In-house shaping cost: \$5–\$12/kg material + \$8–\$20/kg labor = **\$13–\$32/kg finished part** - vs. buying finished soapstone products: \$35–\$80/kg finished

Feasibility Rating: 5/5 — simplest material processing in portfolio. No heat, no chemistry.

Domestic sourcing strategy: Vermont Soapstone Company, Allegheny Gap (Virginia). Build direct quarry relationship for 10+ ton/year volume @ \$4–\$8/kg delivered.

Products: 007 (stone-bake oven), 008 (countertop warmer)

Material 26 — GaN HEMT RF Amplifier Modules (2.45 GHz)

Function: Solid-state microwave source for Safe-Wave microwave oven **Replaces:** Magnetron (1940s vacuum tube technology)

26.1 Performance Specifications

- Frequency: 2.45 GHz $\pm$ 50 MHz
- Output power: 100–900W (single module or combined)
- Drain efficiency: 97% (Class-F topology vs. 65-70% magnetron)
- Voltage: 28–50 V (low-voltage — no 4kV transformer needed)
- Size: 50–100 mm $\times$ 50 mm (small, modular)

26.2 GaN Supply Chain

GaN on SiC wafer (preferred for RF): - GaN epitaxial layer grown by MOCVD on SiC substrate - MOCVD equipment: \$1M–\$3M per reactor (foundry business — not in-house) - Key suppliers: Wolfspeed (Durham NC — domestic!), MACOM, Qorvo, NXP Semiconductors

Commercial GaN RF module pricing: - 100W module at 2.45 GHz: \$45–\$90 each (consumer volume) - 500W combined module: \$180–\$280 - Automotive-qualified: \$250–\$450

Domestic sourcing: Wolfspeed (Cree spin-off) in Durham, NC — US manufacturer. Their expansion facility (Mohawk Valley, NY, 2023) doubles US capacity. **This is a key US-domestic supply chain for CSMFAB product 005.**

26.3 In-House Production: NOT Feasible

GaN RF wafer fabrication requires Class 100 cleanroom + MOCVD + ion implantation. **Strategy: Design modular amplifier PCB using commercial GaN die or modules + custom driver electronics.**

In-house PCB design and assembly: - Design custom driver/controller electronics around commercial GaN module - PCB fabrication (outsourced): \$8–\$25/board - Assembly: \$5–\$15/unit - Total electronics cost: GaN module (\$75) + PCB (\$12) + assembly (\$10) = **\$97/unit electronics package** vs. turnkey GaN microwave module: \$120–\$200

Feasibility Rating: 3/5 — buy GaN RF die/module; design surrounding electronics in-house.

Products: 005 (Safe-Wave microwave)

Material 27 — LoRa Radio Modules (915 MHz)

Technology: Long-Range spread-spectrum radio | **Standard:** LoRa Alliance

27.1 Specifications

- Frequency: 915 MHz (North America), 868 MHz (Europe)
- Range: 2–15 km (urban to line-of-sight)
- Power: 100–250 mW TX
- Data rate: 250 bps – 5.4 kbps
- Sensitivity: -137 dBm (outstanding — reaches through rubble)

27.2 Suppliers and Pricing

Module	Price each (2026)	Form factor
Semtech SX1276 transceiver chip	\$3.80–\$6.00	IC die only
MCCI LoRaWAN module	\$8–\$15	SMD module
Murata LPWA module	\$12–\$20	Certified LPWA
RAK Wireless RAK3172	\$6–\$10	Compact module
Hope RF RFM95W	\$5–\$9	Budget module

At portfolio volume (10,000 units/yr across products 024, 038, 050, CSMMECH 003): Volume pricing: \$4–\$7/module

In-house production: Not feasible (requires IC fabrication). Buy commercial. **Products:** 024, 038, 050, CSMMECH 003

Material 28 — LiFePO4 Battery Cells

Full name: Lithium Iron Phosphate | **Chemistry:** Olivine structure LiFePO4 cathode

28.1 Why LiFePO4 Specifically

- No cobalt (eliminates DRC supply risk)
- No nickel (unlike NMC batteries)
- Thermal runaway temperature: 270°C (vs. 150°C for NMC) — fire-safe
- Cycle life: 3,000–6,000+ cycles (2–4× NMC lifetime)
- Energy density: 90–160 Wh/kg (lower than NMC but sufficient for most Aegis applications)
- Chemical stability in radiation/EM environments: excellent

28.2 Raw Materials

Lithium source: - Spodumene (hard rock, Australia — Pilbara/Greenbushes): Li2O 7% ore → Li2CO3 - Brine (Atacama, Chile/Argentina; Clayton Valley, Nevada USA) - Clay deposits: Thacker Pass (Nevada) — largest US deposit, active permitting - Li2CO3 price 2026: \$12–\$22/kg (after 2022–2023 peak of \$75/kg)

Iron phosphate (FePO4): - From iron ore + phosphoric acid (phosphate rock) — both abundant and cheap - FePO4 cost: \$2–\$4/kg

Cell assembly: - Cathode: LiFePO4 + carbon black + PVDF binder on aluminum foil - Anode: graphite + PVDF binder on copper foil - Separator: polyethylene microporous membrane - Electrolyte: LiPF6 in organic carbonate solvent

28.3 Commercial Pricing

Format	Price (2026)	Supplier
LiFePO4 cells (cylindrical 32700)	\$0.90–\$1.30/Wh	Eve Energy, CATL
LiFePO4 cells (prismatic)	\$0.85–\$1.20/Wh	BYD, CATL, Gotion
LiFePO4 pouch	\$1.10–\$1.60/Wh	Amprius, Farasis
Domestic US (Ucore Rare Metals)	\$1.80–\$2.50/Wh	Premium for domestic

Portfolio battery demand: - Product 019 (forklift): 40 kWh □ \$8,500–\$16,000 battery pack - Product 050 (drone): 5.4 kWh □ \$1,150–\$2,160 - Products 024, 038 (micro-batteries): 30 mAh each □ \$0.03–\$0.05 each

28.4 In-House Battery Production: Future Consideration

At portfolio scale, battery assembly (not cell manufacturing) becomes viable: - Cell procurement + module assembly + BMS design - Equipment: automated cell stacker + spot welder + BMS circuit board - Investment: \$80,000–\$200,000 for pack assembly line - Benefit: 30-40% cost reduction vs. buying finished packs

Feasibility Rating: 3/5 — cell assembly makes sense at scale; cell production requires >\$500M.

Material 29 — PMMA Plastic Optical Fiber (POF)

Material: Polymethyl methacrylate (same as Plexiglas/acrylic) | **Core diameter:** 0.5–2 mm **Function:** Signal transmission in wiring systems; replaces copper signal wires for GIC immunity

29.1 Properties

- Optical bandwidth: 100 MHz·km (adequate for all sensor and control signals)
- Attenuation: 200 dB/km at 660 nm (significant — limits length to <100m without repeaters)
- Temperature range: -40°C to +85°C (adequate for vehicle and building applications)
- Bend radius: 25 mm minimum (flexible, easy installation)
- **Zero EM susceptibility — immune to all GIC, EMP, EMF**

29.2 Commercial Pricing

- Mitsubishi Rayon Eska (Japan): \$0.45–\$0.80/m
- Industrial Fiber Optics (USA): \$0.55–\$1.20/m
- Polymer Optics (UK): specialty \$1.50–\$4.00/m
- Bulk cable (1000m spools): \$0.25–\$0.50/m

At portfolio volume (50 km/year estimate): \$0.22–\$0.40/m supply agreement

29.3 In-House Drawing of PMMA Fiber

PMMA fiber drawing from MMA-grade PMMA rod □ heated draw tower Equipment: fiber draw tower (1–2m tower): \$15,000–\$35,000 In-house including MMA polymer: **\$0.08–\$0.15/m vs. \$0.22–\$0.80 commercial Savings: 55–80% Feasibility Rating: 4/5** — fiber drawing is low-complexity; MMA/PMMA readily available.

Material 30 — Piezoelectric Materials: KNbO₃-BaTiO₃ and KNbO₃

Replacing: PZT (lead zirconate titanate) — legally restricted under RoHS Annex II (EU) and increasingly regulated in US

30.1 KNbO₃-BaTiO₃ Properties (V2.0 Specification)

- d₃₃: 285 pC/N (comparable to PZT-5H at 289 pC/N)
- Curie temperature T_c: 310°C (vs. 193°C for PZT-5H — 117°C higher thermal margin)
- Lead-free, RoHS/REACH compliant
- Dielectric constant: 600–1200

30.2 Raw Material Precursors

Potassium (K₂CO₃): Sylvite (KCl) mining □ potash industry; \$0.60/kg **Niobium (Nb₂O₅):** Columbite-tantalite (coltan) ore from Brazil/DRC; \$30–\$55/kg Nb₂O₅ **Barium (BaCO₃):** Witherite mining or chemical precipitation; \$2.50/kg **Titanium (TiO₂):** Rutile/ilmenite mining; \$3–\$8/kg

Synthesis route (solid-state reaction): K₂CO₃ + Nb₂O₅ □ 2KNbO₃ (1000°C, 4h, in air) xKNbO₃ + (1-x)BaTiO₃ □ solid solution □ sinter to dense ceramics or press to piezo elements

In-house synthesis cost: Nb₂O₅ (\$40/kg) × 0.62 kg/kg KNbO₃ + other components = ~\$28–\$40/kg KNbO₃-BaTiO₃ powder Commercial: \$180–\$250/kg **In-house savings:** 75–85% **Feasibility Rating:** 4/5 — standard solid-state synthesis; shared with ceramics program.

Products: Virtually all Embark products + stove, oven, water heater, plumbing, robotics

Summary Table — Part IV Polymers, Electronics, Support

Material	Commercial Price	In-House Path	In-House Cost	Rating
PEEK CF40	\$220–\$320/kg	Machine commercial stock	N/A (buy)	3/5
PEX-a pipe	\$2.80–\$6.20/m	Extrusion at volume	viable >100km/yr	2/5
PEN spray can	\$3.50–\$6.50/kg pellet	ISBM blow molding	\$0.65–\$0.95/can	4/5
Soapstone	\$2–\$15/kg	Quarry + shape in-house	\$13–\$32/kg finished	5/5
GaN RF module	\$45–\$90/module	Design PCB around module	\$97/electronics pkg	3/5
LoRa radio	\$5–\$20/module	Buy commercial	N/A	1/5
LiFePO ₄ cells	\$0.85–\$2.50/Wh	Pack assembly in-house	30-40% savings at scale	3/5
PMMA optical fiber	\$0.22–\$0.80/m	In-house draw tower	\$0.08–\$0.15/m	4/5
KNbO ₃ -BaTiO ₃ piezo	\$180–\$250/kg	Solid-state synthesis	\$28–\$40/kg	4/5

End of CSMFAB Materials Study Part IV | Polymers, Electronics, and Support Materials Carrington Storm Motors | June 2026